

Citrix Hypervisor Performance Tips



JulienDestombes October 13, 2020 in [Integration](#)

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JulienDestombes Members
October 13, 2020



Hi all,

Few month ago I post this thread <https://discussions.citrix.com/topic/407437-citrix-hypervisor-performance-tuning/#comment-2071495>

Quote

Hi,

I need to configure Citrix Hypervisor 8.1 on HPe Synergy 480 Gen10 for XenApp Workload (no GPU).

I found HPe whitepaper on BIOS tuning for XenApp workload

<https://h20195.www2.hpe.com/v2/getdocument.aspx?docname=a00049199enw>

Can I also apply those settings to Citrix Hypervisor ? For instance, the whitepaper disable C-State but I read an excellent post of Tobias that says to enable C-States.

true ▼

Now we are on migration and we observe that CPU consumption is very high (30 users take 90% of a 8 vCPUs VMs and 40% of the host CPU with only one VM on it).

The CPU are 2 x Intel Xeon-Gold 6244 (3.6GHz/8-core/150W) so we use VMs (4 by host) with 8 vCPUs and all Citrix optimization from Citrix Optimizer.

WEM is also in place.

BIOS is configured like this:

Power Management (Power Profile): Maximum Performance

Power Management (Power Regulator): Static High Performance Mode (Default)

Minimum Processor Idle Power Core C-State: No C-states

Minimum Processor Idle Package C-State: No Package States

Intel QPI Link Power Management: Disabled

Energy Performance Bios: Maximum Performance

QPI Bandwidth Optimization (RTID): Optimized for I/O (Alternate RTID)

Hyper-Threading: Enabled

Intel Turbo Boost: Enabled

What is your suggestion to improve performances ? I don't know if we can improve something but I am quite disappointing to see that one VM take 40% of CPU of my host.

Thanks for helping me to troubleshoot this issue :)

+ “ Quote



Tobias Kreidl Members
October 13, 2020



What sort of storage do you have? How is it configured? Now many VMs per SR do you have? Also, whats your dom0 memory set to use? Run top and xentop from the CLI and look for bottlenecks, like CPU saturation, swapping, etc. Also run iostat and look for disk I/O bottlenecks like high await and read/write times or saturated queues.

There are so many possible reasons for this, it's not just a simple do this or do that to fix it type of situation. People spend years studying how to optimize performance.

--Tobias

+  Quote



JulienDestombes

Members

October 13, 2020

AUTHOR



Hi Tobias,

I know there is not a simple answer to advise to me: tick this option and everything will be great :6

But your advises to check xentop, swapping and do iostat are great ideas. The customer pin me about this problem and I go on site on Thursday so I try to collect everything I can to gain time.

To answer your questions:

1-What sort of storage do you have?

We use SAN storage

2-How is it configured?

We use fiber channel

3-Now many VMs per SR do you have?

We are at the beginning of the deployment so no more than 8 VMs are deployed on the SR with MCS

4-Also, whats your dom0 memory set to use?

Dom0 was configured with default settings (so it should be 8 GiB according to Citrix docs "By default, Citrix Hypervisor allocates 1 GiB plus 5% of the total physical memory to the control domain, up to an initial maximum of 8 GiB"; we have 512 GiB on our servers)

domain, up to an initial maximum of 8 GiB , we have 312 GiB on our servers).

Thanks very much for the time you take to answer me. I really appreciate.

Regards

Julien

+  Quote



Tobias Kreidl Members

October 13, 2020



Julien,

Make sure to check storage I/O stats. This is often one of the most overlooked areas and it is critical to good performance. With that small a number of VMs per SR, the queue depth should not be an issue. Consider also various read cache options, in particular being now on 8.X. We used a NexentaStor array that with our provisioned MCS VMs, achieved a very high cache read hit rate and because it was all SSD drives, reads were super fast in any case. With iostats, any delays in storage I/O over around 20 to 30 msec will start to be noticeable.

--Tobias

+  Quote



Christof Giesers Legacy Group

October 13, 2020



“ Quote

Power Management (Power Profile): Maximum Performance

Power Management (Power Regulator): Static High Performance Mode (Default)

Isn't it recommended (for a while now) to set it to OS?

Edit: Go to console (exit to shell) and use top to see if it's io-waits or how dom0 behaves.

(Press 1 to see it for each core you assigned to dom0)

[+](#) [“ Quote](#)

**JulienDestombes**

Members

October 14, 2020

[AUTHOR](#)

Hi Christof,

I know that it's recommended by this kb <https://support.citrix.com/article/CTX200390>

But I follow an HP white paper (based on vSphere) to configure BIOS for XenApp Workload (you can find this white paper on the reference architecture from TechZone).

I will follow your tips to check io-waits.

Thanks !

+ “ Quote



Tobias Kreidl Members

October 14, 2020



For storage, the iowait and access times are important. Overall performance is not just about the CPUs. As to the CPU, whether or not to set to maximum performance depends on other factors, such as if GPUs are being used or not. Enabling C-states is generally recommended. See for example <https://www.mycugc.org/blogs/tobias-kreidl/2019/03/07/tale-of-two-servers-bios-settings-affect-apps-gpu> and the follow-up two articles <https://www.mycugc.org/blogs/tobias-kreidl/2019/04/30/a-tale-of-two-servers-part-2> and <https://www.mycugc.org/blogs/tobias-kreidl/2019/04/30/a-tale-of-two-servers-part-3>. Also, when configuring large VMs, NUMA/vNUMA should be taken into consideration, in particular for older CPU architectures.

--Tobias

+ “ Quote



sjoerdvbn Members
October 14, 2020



Hi,

For HPE servers look at Jitter Control. We had an issue with our customer where we also had some performance issue and we set Jitter Control to manual and the "Processor Jitter Control Frequency(in MHz)" to the base CPU speed and we've got a stable environment.

https://support.hpe.com/hpesc/public/docDisplay?docId=a00018313en_us

Regards.

+ “ Quote



JulienDestombes Members
October 14, 2020

[AUTHOR](#)



Hi,

Thanks for the tips !

My colleague is on customer site tomorrow, I will ask him to test it !

I 'll keep you informed !

Thanks a lot to everybody that try to help me, I really appreciate.

+  Quote



Christof Giesers Legacy Group
October 14, 2020



Right, I remember that if you disable all "savings features", e.g. C-states etc. it can't utilize the turbo states as other cores need to scale down to let a few scale up.

+  Quote



Tobias Kreidl Members
October 14, 2020



“



Right, I remember that if you disable all "savings features", e.g. C-states etc. it can't utilize the turbo states as other cores need to scale down to let a few scale up.

Indeed -you don't want to disable C-states, ever. Turbo mode is a matter of a whole separate discussion - see the references to my articles above, for example.

-=Tobias



“ Quote



JulienDestombes

Members

October 19, 2020

[AUTHOR](#)



Hi All,

For information, we have applied recommandation for Jitter Control (not sure that helps for us).

And we also applied <https://support.citrix.com/article/CTX269317>. This one seems to help (we have to test on more VMs to be sure).

From my perspective, iowaits, top and xentop return no problem.

I will keep you inform.

Julien

+ “ Quote

3 weeks later...



JulienDestombes

Members

November 7, 2020

AUTHOR



Hi all,

I want to share you my experience return.

We try different BIOS config and for my perspective OS control and turbo mode give a good improvement.

However I want to try another config to disable C-State as I see on this blog it could impact CPU usage

<https://www.wedelit.no/blog/2018/01/05/xenserver-c-state-or-not-to-c-state/>

I was thinking that C-States was disabled, I checked BIOS and it is but if I run

```
xenpm get-cpuidle-states | grep total | uniq
```

I still see 4 and not 2 as result. Is there a file or something to do to update CPU info ?

Thanks for your help.

Julien

+  Quote



JulienDestombes

Members

November 12, 2020

[AUTHOR](#)



Hi,

I found the command line:

```
/opt/xensource/libexec/xen-cmdline --set-xen mwait-idle=false
```

Regards

+  Quote



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